EN3160 Image Processing and Machine Vision

Assignment 3 – Neural Networks

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GitHub: denethp/CV-Assignment-03

Question 1

[2]:
```
import torch
import torch.nn as nn
import torchvision
import torchvision.transforms as transforms
import matplotlib.pyplot as plt

%matplotlib inline

_ = torch.manual_seed(0)  # fixed seed so my runs are reproducible
```

[7]:
```
# CIFAR-10, normalized to [-1, 1]
transform = transforms.Compose(
    [transforms.ToTensor(),
     transforms.Normalize((0.5, 0.5, 0.5), (0.5, 0.5, 0.5))])

batch_size = 50

trainset = torchvision.datasets.CIFAR10(root='./data', train=True,
                                        download=True, transform=transform)
trainloader = torch.utils.data.DataLoader(trainset, batch_size=batch_size,
                                          shuffle=True, num_workers=2)

testset = torchvision.datasets.CIFAR10(root='./data', train=False,
                                       download=True, transform=transform)
testloader = torch.utils.data.DataLoader(testset, batch_size=batch_size,
                                         shuffle=False, num_workers=2)

classes = ('plane', 'car', 'bird', 'cat', 'deer',
           'dog', 'frog', 'horse', 'ship', 'truck')
```

[8]:
```
def accuracy(loader, forward):
    # Percentage of correctly classified images in a loader
    correct, total = 0, 0
    with torch.no_grad():
        for inputs, labels in loader:
            x = inputs.view(inputs.shape[0], -1)
            predicted = torch.argmax(forward(x), dim=1)
            total += labels.size(0)
            correct += (predicted == labels).sum().item()
    return 100 * correct / total
```

Baseline: single dense layer with MSE loss

Before making the required changes, I first trained the given network as it is, so I have a reference to compare against. It maps the flattened image ($3 \times 32 \times 32 = 3072$ values) straight to the 10 class scores with a single weight matrix and is trained with a mean-squared-error loss.

[10]:
```
Din = 3 * 32 * 32  # flattened CIFAR-10 image
K = 10             # number of classes
std = 1e-5

w = torch.randn(Din, K) * std
b = torch.zeros(K)

iterations = 20
lr = 2e-6
lr_decay = 0.9
reg = 0
loss_history = []

for t in range(iterations):
    running_loss = 0.0
    for inputs, labels in trainloader:
        Ntr = inputs.shape[0]
        x_train = inputs.view(Ntr, -1)
        y_train_onehot = nn.functional.one_hot(labels, K).float()

        # Forward pass
        y_pred = x_train.mm(w) + b

        # MSE loss with L2 regularization
        loss = (1 / Ntr) * torch.sum((y_pred - y_train_onehot) ** 2) + reg * torch.sum(w ** 2)
        loss_history.append(loss.item())
        running_loss += loss.item()

        # Backpropagation
        dy_pred = (2.0 / Ntr) * (y_pred - y_train_onehot)
        dw = x_train.t().mm(dy_pred) + reg * w
        db = dy_pred.sum(dim=0)

        w -= lr * dw
        b -= lr * db

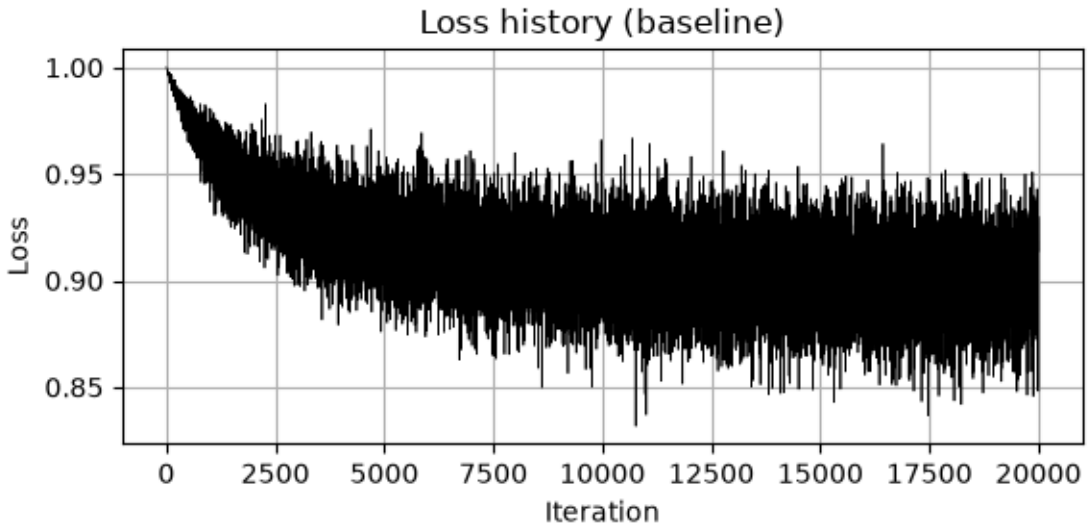
    print(f'Epoch {t + 1}/{iterations}, Loss: {running_loss / len(trainloader):.4f}')
    lr *= lr_decay
```

```
Epoch 1/20, Loss: 0.9770
Epoch 2/20, Loss: 0.9499
Epoch 3/20, Loss: 0.9361
Epoch 4/20, Loss: 0.9275
Epoch 5/20, Loss: 0.9216
Epoch 6/20, Loss: 0.9172
Epoch 7/20, Loss: 0.9138
Epoch 8/20, Loss: 0.9111
Epoch 9/20, Loss: 0.9088
Epoch 10/20, Loss: 0.9070
Epoch 11/20, Loss: 0.9054
Epoch 12/20, Loss: 0.9041
Epoch 13/20, Loss: 0.9029
Epoch 14/20, Loss: 0.9020
Epoch 15/20, Loss: 0.9011
Epoch 16/20, Loss: 0.9004
Epoch 17/20, Loss: 0.8997
Epoch 18/20, Loss: 0.8991
Epoch 19/20, Loss: 0.8986
Epoch 20/20, Loss: 0.8982
```

[11]:
```
baseline_train_acc = accuracy(trainloader, lambda x: x.mm(w) + b)
baseline_test_acc = accuracy(testloader, lambda x: x.mm(w) + b)
print(f'Training accuracy: {baseline_train_acc:.2f}%')
print(f'Test accuracy: {baseline_test_acc:.2f}%')

plt.figure(figsize=(6, 3))
plt.plot(loss_history, color='k', linewidth=0.5)
plt.title('Loss history (baseline)')
plt.xlabel('Iteration')
plt.ylabel('Loss')
plt.grid(True)
plt.tight_layout()
plt.show()
```

```
Training accuracy: 32.23%
Test accuracy: 32.43%
```



(a) Adding a middle layer with 100 nodes and sigmoid activation

I added a hidden layer of $H = 100$ nodes between the input and the output, so the network becomes

$$\mathbf{h} = \sigma(\mathbf{x}W_1 + \mathbf{b}_1), \qquad \hat{\mathbf{y}} = \mathbf{h}W_2 + \mathbf{b}_2,$$

with $W_1 \in \mathbb{R}^{3072\times100}$ and $W_2 \in \mathbb{R}^{100\times10}$. I still compute the forward pass and the gradients by hand. Going backwards from the output gradient $\partial L/\partial \hat{\mathbf{y}}$:

$$\frac{\partial L}{\partial W_2} = \mathbf{h}^T \frac{\partial L}{\partial \hat{\mathbf{y}}}, \qquad \frac{\partial L}{\partial \mathbf{z}_1} = \left(\frac{\partial L}{\partial \hat{\mathbf{y}}} W_2^T\right) \odot \mathbf{h} \odot (1 - \mathbf{h}), \qquad \frac{\partial L}{\partial W_1} = \mathbf{x}^T \frac{\partial L}{\partial \mathbf{z}_1},$$

where $\mathbf{h} \odot (1 - \mathbf{h})$ is the derivative of the sigmoid. I verified these gradients against numerical (finite-difference) gradients before training.

The tiny initialization (10^{-5}) of the single-layer network does not work with a hidden layer, because every hidden node would output $\sigma(0) = 0.5$ and the signal would barely reach the output. So I initialized each weight matrix with standard deviation $1/\sqrt{\text{fan-in}}$. To speed up training, I loaded the whole dataset onto the GPU once and drew the mini-batches from it directly.

[12]:
```
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print('Device:', device)

def to_device_tensors(dataset):
    # Same preprocessing as the transform above: [0, 1] -> [-1, 1], then flatten (C, H, W)
    x = torch.tensor(dataset.data, dtype=torch.float32).permute(0, 3, 1, 2) / 255.0
    x = ((x - 0.5) / 0.5).reshape(len(dataset), -1)
    y = torch.tensor(dataset.targets)
    return x.to(device), y.to(device)

X_train, Y_train = to_device_tensors(trainset)
X_test, Y_test = to_device_tensors(testset)
print('Train:', tuple(X_train.shape), ' Test:', tuple(X_test.shape))
```

```
Device: cuda
Train: (50000, 3072)  Test: (10000, 3072)
```

[13]:
```
Din = 3 * 32 * 32
H = 100  # hidden nodes
K = 10

def init_params(seed=0):
    g = torch.Generator(device=device).manual_seed(seed)
    w1 = torch.randn(Din, H, device=device, generator=g) / Din ** 0.5
    b1 = torch.zeros(H, device=device)
    w2 = torch.randn(H, K, device=device, generator=g) / H ** 0.5
    b2 = torch.zeros(K, device=device)
    return [w1, b1, w2, b2]

def forward(x, params):
    w1, b1, w2, b2 = params
    h = torch.sigmoid(x.mm(w1) + b1)  # hidden layer
    y_pred = h.mm(w2) + b2            # output scores
    return y_pred, h

def evaluate(X, Y, params):
    with torch.no_grad():
        predicted = torch.argmax(forward(X, params)[0], dim=1)
    return 100 * (predicted == Y).float().mean().item()

def train(loss_fn, epochs=10, lr=0.05, lr_decay=0.9, reg=0.0, seed=0):
    # Mini-batch gradient descent with manually computed gradients
    params = init_params(seed)
    w1, b1, w2, b2 = params
    g = torch.Generator(device=device).manual_seed(seed)
    N = X_train.shape[0]
    loss_history, epoch_loss = [], []

    for epoch in range(epochs):
        perm = torch.randperm(N, device=device, generator=g)
        running_loss = 0.0
        for i in range(0, N, batch_size):
            idx = perm[i:i + batch_size]
```

```
            x = X_train[idx]
            y_onehot = nn.functional.one_hot(Y_train[idx], K).float()

            # Forward pass
            y_pred, h = forward(x, params)
            loss, dy_pred = loss_fn(y_pred, y_onehot)
            loss = loss + reg * (torch.sum(w1 ** 2) + torch.sum(w2 ** 2))

            # Backpropagation
            dw2 = h.t().mm(dy_pred) + 2 * reg * w2
            db2 = dy_pred.sum(dim=0)
            dz1 = dy_pred.mm(w2.t()) * h * (1 - h)  # through the sigmoid
            dw1 = x.t().mm(dz1) + 2 * reg * w1
            db1 = dz1.sum(dim=0)

            # Parameter update
            w1 -= lr * dw1
            b1 -= lr * db1
            w2 -= lr * dw2
            b2 -= lr * db2

            loss_history.append(loss.item())
            running_loss += loss.item()

        epoch_loss.append(running_loss / (N // batch_size))
        print(f'Epoch {epoch + 1}/{epochs}, Loss: {epoch_loss[-1]:.4f}')
        lr *= lr_decay

    return params, loss_history, epoch_loss
```

To see the effect of the hidden layer on its own, I first trained it with the same MSE loss as the baseline, for 10 epochs.

[14]:
```
def mse_loss(y_pred, y_onehot):
    n = y_pred.shape[0]
    loss = torch.sum((y_pred - y_onehot) ** 2) / n
    dy_pred = 2.0 * (y_pred - y_onehot) / n
    return loss, dy_pred

params_mse, loss_history_mse, epoch_loss_mse = train(mse_loss, epochs=10, lr=0.05)

mse_train_acc = evaluate(X_train, Y_train, params_mse)
mse_test_acc = evaluate(X_test, Y_test, params_mse)
print(f'Training accuracy: {mse_train_acc:.2f}%')
print(f'Test accuracy: {mse_test_acc:.2f}%')
```

```
Epoch 1/10, Loss: 2.0829
Epoch 2/10, Loss: 0.8287
Epoch 3/10, Loss: 0.7952
Epoch 4/10, Loss: 0.7732
Epoch 5/10, Loss: 0.7589
Epoch 6/10, Loss: 0.7472
Epoch 7/10, Loss: 0.7385
Epoch 8/10, Loss: 0.7321
Epoch 9/10, Loss: 0.7258
Epoch 10/10, Loss: 0.7208
Training accuracy: 47.71%
Test accuracy: 44.01%
```

(b) Using the cross-entropy loss

Next, I replaced the MSE loss with the cross-entropy loss. The output scores $\hat{\mathbf{y}}$ are first turned into class probabilities with the softmax function, and the loss is the average negative log-probability of the correct class:

$$p_k = \frac{e^{\hat{y}_k}}{\sum_{j=1}^{K} e^{\hat{y}_j}}, \qquad L = -\frac{1}{N}\sum_{i=1}^{N}\sum_{k=1}^{K} y_{ik} \log p_{ik}.$$

The gradient of this loss with respect to the output scores simplifies to

$$\frac{\partial L}{\partial \hat{\mathbf{y}}} = \frac{1}{N}(\mathbf{p} - \mathbf{y}),$$

so only the output gradient changes, and the rest of my backpropagation through the hidden layer stays the same. I computed the softmax through `log_softmax`, which subtracts the largest score first, so that large scores cannot overflow the exponential. I also checked this gradient numerically, and kept every other setting identical to the MSE run (10 epochs, learning rate 0.05, decay 0.9) so that the two losses can be compared fairly.

[15]:
```
def cross_entropy_loss(y_pred, y_onehot):
    n = y_pred.shape[0]
    log_p = torch.log_softmax(y_pred, dim=1)  # numerically stable log-softmax
    loss = -torch.sum(y_onehot * log_p) / n
    dy_pred = (torch.exp(log_p) - y_onehot) / n
    return loss, dy_pred

params_ce, loss_history_ce, epoch_loss_ce = train(cross_entropy_loss, epochs=10, lr=0.05)

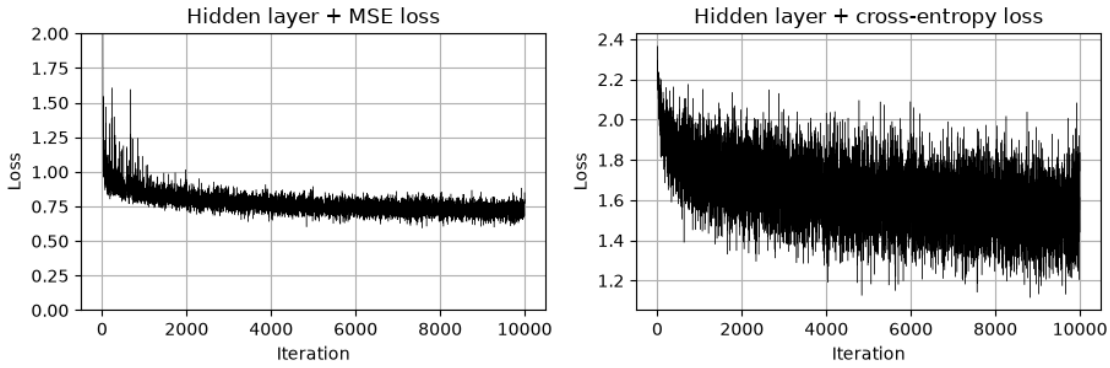
ce_train_acc = evaluate(X_train, Y_train, params_ce)
ce_test_acc = evaluate(X_test, Y_test, params_ce)
print(f'Training accuracy: {ce_train_acc:.2f}%')
print(f'Test accuracy: {ce_test_acc:.2f}%')
```

```
Epoch 1/10, Loss: 1.8510
Epoch 2/10, Loss: 1.7234
Epoch 3/10, Loss: 1.6740
Epoch 4/10, Loss: 1.6395
Epoch 5/10, Loss: 1.6123
Epoch 6/10, Loss: 1.5899
Epoch 7/10, Loss: 1.5693
Epoch 8/10, Loss: 1.5524
Epoch 9/10, Loss: 1.5369
Epoch 10/10, Loss: 1.5223
Training accuracy: 48.55%
Test accuracy: 45.94%
```

(c) Results after 10 epochs

The final network (100-node sigmoid hidden layer with the cross-entropy loss) was trained for 10 epochs in part (b). Below I plot its loss next to the MSE run and compare the accuracies of all three networks I trained.

[16]:
```
fig, axes = plt.subplots(1, 2, figsize=(9.3, 3.2))
curves = [(loss_history_mse, 'Hidden layer + MSE loss'),
          (loss_history_ce, 'Hidden layer + cross-entropy loss')]
for ax, (hist, title) in zip(axes, curves):
    ax.plot(hist, color='k', linewidth=0.3)
    ax.set_title(title)
    ax.set_xlabel('Iteration')
    ax.set_ylabel('Loss')
    ax.grid(True)
axes[0].set_ylim(0, 2)  # the first few MSE iterations are much larger
plt.tight_layout()
plt.show()
```



[17]:
```
results = [('Single layer + MSE (20 epochs)', baseline_train_acc, baseline_test_acc),
           ('Hidden layer + MSE (10 epochs)', mse_train_acc, mse_test_acc),
           ('Hidden layer + cross-entropy (10 epochs)', ce_train_acc, ce_test_acc)]

print(f"{'Network':<42}{'Train acc':>11}{'Test acc':>11}")
for name, tr, te in results:
    print(f'{name:<42}{tr:>10.2f}%{te:>10.2f}%')
```

```
Network                                    Train acc   Test acc
Single layer + MSE (20 epochs)               32.23%     32.43%
Hidden layer + MSE (10 epochs)               47.71%     44.01%
Hidden layer + cross-entropy (10 epochs)     48.55%     45.94%
```

Discussion: adding the 100-node sigmoid hidden layer gave the biggest improvement. With the same MSE loss, the test accuracy rose from 32.43% (single layer, 20 epochs) to 44.01% in only 10 epochs. The single-layer network is a linear classifier: it can only learn one template per class, and its training and test accuracies were almost equal (32.23% vs 32.43%), which shows it was limited by its capacity rather than by overfitting. The sigmoid hidden layer lets the network learn non-linear combinations of pixels, so it fits the data much better.

Switching from MSE to cross-entropy, with every other setting unchanged, improved the test accuracy further to 45.94% (training 48.55%). I think this is because the cross-entropy gradient $(\mathbf{p}-\mathbf{y})/N$ keeps pushing the score of the correct class up until it dominates through the softmax, whereas MSE forces all ten outputs towards exact 0/1 targets and spends effort on the wrong classes' scores even when the prediction is already correct. The loss curves also show this: the MSE run had a large spike in the first iterations before settling, while the cross-entropy loss fell

smoothly from the start. The gap between training and test accuracy also became smaller with cross-entropy (2.6 points vs 3.7 points for MSE).

The cross-entropy loss was still decreasing at the end of the 10 epochs, so training for longer (or with a larger learning rate) would likely raise the accuracy further. However, a fully connected network on raw pixels ignores the spatial structure of the image, so it cannot get close to the accuracy of a convolutional network on CIFAR-10.

Question 2

[3]:
```
import time
import torch.nn.functional as F
import torch.optim as optim

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')

# MNIST, normalized with the dataset mean and standard deviation
mnist_train = torchvision.datasets.MNIST(root='./data', train=True, download=True)
mnist_test = torchvision.datasets.MNIST(root='./data', train=False, download=True)

def mnist_to_device(dataset):
    x = dataset.data.float().unsqueeze(1) / 255.0  # (N, 1, 28, 28)
    x = (x - 0.1307) / 0.3081
    return x.to(device), dataset.targets.to(device)

Xm_train, Ym_train = mnist_to_device(mnist_train)
Xm_test, Ym_test = mnist_to_device(mnist_test)
print('Train:', tuple(Xm_train.shape), ' Test:', tuple(Xm_test.shape))
```

```
Train: (60000, 1, 28, 28)  Test: (10000, 1, 28, 28)
```

To train both networks under exactly the same conditions, I wrote one training function and use it for both models. It uses the Adam optimizer (learning rate 10^{-3}), mini-batches of 64 images and the cross-entropy loss, and trains for 10 epochs. After every epoch it records the training and test loss and accuracy, so that I can plot the learning curves later. The training values are averaged over the epoch while the model is in training mode, and the test values are computed in evaluation mode. As in Question 1, I kept the whole dataset on the GPU and drew the batches from it directly.

[4]:
```
def evaluate_model(model, X, Y, batch=1000):
    # Average loss and accuracy (%) in evaluation mode
    model.eval()
    total_loss, correct = 0.0, 0
    with torch.no_grad():
        for i in range(0, len(X), batch):
            out = model(X[i:i + batch])
            total_loss += F.cross_entropy(out, Y[i:i + batch], reduction='sum').item()
            correct += (out.argmax(dim=1) == Y[i:i + batch]).sum().item()
    return total_loss / len(X), 100 * correct / len(X)

def fit(model, epochs=10, lr=1e-3, batch=64, seed=0):
    torch.manual_seed(seed)
    optimizer = optim.Adam(model.parameters(), lr=lr)
    history = {'train_loss': [], 'train_acc': [], 'test_loss': [], 'test_acc': []}
    N = len(Xm_train)

    for epoch in range(epochs):
        model.train()
```

```
        start = time.time()
        perm = torch.randperm(N, device=device)
        total_loss, correct = 0.0, 0
        for i in range(0, N, batch):
            idx = perm[i:i + batch]
            x, y = Xm_train[idx], Ym_train[idx]

            out = model(x)
            loss = F.cross_entropy(out, y)
            optimizer.zero_grad()
            loss.backward()
            optimizer.step()

            total_loss += loss.item() * len(y)
            correct += (out.argmax(dim=1) == y).sum().item()

        test_loss, test_acc = evaluate_model(model, Xm_test, Ym_test)
        history['train_loss'].append(total_loss / N)
        history['train_acc'].append(100 * correct / N)
        history['test_loss'].append(test_loss)
        history['test_acc'].append(test_acc)
        print(f"Epoch {epoch + 1}/{epochs}: "
              f"train loss {history['train_loss'][-1]:.4f}, train acc {history['train_acc'][-1]:.
↪2f}% | "
              f"test loss {test_loss:.4f}, test acc {test_acc:.2f}% ({time.time() - start:.
↪1f}s)")
    return history
```

(a) LeNet-5

I built LeNet-5 following the original design by LeCun et al. (1998). Since MNIST images are 28×28 rather than the 32×32 inputs of the original network, I padded the first convolution by 2 pixels so that the feature map sizes are the same as in the paper:

- C1: 5×5 convolution with 6 filters and padding 2, tanh $\rightarrow 6 \times 28 \times 28$
- S2: 2×2 average pooling $\rightarrow 6 \times 14 \times 14$
- C3: 5×5 convolution with 16 filters, tanh $\rightarrow 16 \times 10 \times 10$
- S4: 2×2 average pooling $\rightarrow 16 \times 5 \times 5$
- C5: fully connected, 120 units, tanh
- F6: fully connected, 84 units, tanh
- Output: fully connected, 10 units (one score per digit)

I kept the original tanh activations and average pooling, so that this network represents the classic design and the modern network in part (b) shows the effect of newer techniques.

[5]:
```
class LeNet5(nn.Module):
    def __init__(self):
        super().__init__()
        self.c1 = nn.Conv2d(1, 6, kernel_size=5, padding=2)
        self.s2 = nn.AvgPool2d(2)
        self.c3 = nn.Conv2d(6, 16, kernel_size=5)
        self.s4 = nn.AvgPool2d(2)
        self.c5 = nn.Linear(16 * 5 * 5, 120)
        self.f6 = nn.Linear(120, 84)
        self.out = nn.Linear(84, 10)
```

```
    def forward(self, x):
        x = self.s2(torch.tanh(self.c1(x)))
        x = self.s4(torch.tanh(self.c3(x)))
        x = torch.flatten(x, 1)
        x = torch.tanh(self.c5(x))
        x = torch.tanh(self.f6(x))
        return self.out(x)

torch.manual_seed(0)
lenet = LeNet5().to(device)
print(f'LeNet-5 parameters: {sum(p.numel() for p in lenet.parameters()):,}')

history_lenet = fit(lenet, epochs=10)
print(f"\nLeNet-5 after 10 epochs: training accuracy {history_lenet['train_acc'][-1]:.2f}%, "
      f"test accuracy {history_lenet['test_acc'][-1]:.2f}%")
```

```
LeNet-5 parameters: 61,706
Epoch 1/10: train loss 0.2575, train acc 92.54% | test loss 0.0886, test acc
97.55% (6.3s)
Epoch 2/10: train loss 0.0735, train acc 97.81% | test loss 0.0607, test acc
98.09% (5.0s)
Epoch 3/10: train loss 0.0524, train acc 98.36% | test loss 0.0526, test acc
98.27% (4.7s)
Epoch 4/10: train loss 0.0396, train acc 98.77% | test loss 0.0542, test acc
98.28% (4.4s)
Epoch 5/10: train loss 0.0334, train acc 98.95% | test loss 0.0458, test acc
98.39% (4.5s)
Epoch 6/10: train loss 0.0277, train acc 99.12% | test loss 0.0463, test acc
98.64% (4.7s)
Epoch 7/10: train loss 0.0242, train acc 99.21% | test loss 0.0380, test acc
98.71% (4.6s)
Epoch 8/10: train loss 0.0205, train acc 99.35% | test loss 0.0491, test acc
98.57% (4.7s)
Epoch 9/10: train loss 0.0170, train acc 99.46% | test loss 0.0468, test acc
98.56% (4.6s)
Epoch 10/10: train loss 0.0148, train acc 99.50% | test loss 0.0380, test acc
98.80% (4.8s)

LeNet-5 after 10 epochs: training accuracy 99.50%, test accuracy 98.80%
```

(b) A modern small CNN with batch normalization and dropout

For the modern network I used three convolutional blocks. Each block is a 3×3 convolution, batch normalization, a ReLU activation and 2×2 max pooling, and the number of filters doubles from block to block:

- Block 1: 32 filters $\rightarrow 32 \times 14 \times 14$
- Block 2: 64 filters $\rightarrow 64 \times 7 \times 7$
- Block 3: 128 filters $\rightarrow 128 \times 3 \times 3$
- Classifier: dropout (0.5), fully connected layer with 128 units and ReLU, dropout (0.5), and a final layer with 10 outputs

Compared with LeNet-5, I made the following changes:

- ReLU instead of tanh, which does not saturate, so the gradients stay large in the early layers.
- Batch normalization after every convolution, which keeps the input to each layer normalized during training. This makes training faster and more stable. The convolutions have no bias

because batch normalization adds its own shift.

- Max pooling instead of average pooling, which keeps the strongest response in each region.
- Dropout in the classifier, which randomly switches off half of the units during training so that the network cannot rely on a few specific features. This acts as a regularizer against overfitting.
- Three convolutional layers with more filters, so it can learn richer features.

I trained it with exactly the same `fit` function and settings as LeNet-5 (Adam, learning rate 10^{-3}, batch size 64, 10 epochs).

[6]:
```
class ModernCNN(nn.Module):
    def __init__(self):
        super().__init__()
        def block(c_in, c_out):
            return nn.Sequential(
                nn.Conv2d(c_in, c_out, kernel_size=3, padding=1, bias=False),
                nn.BatchNorm2d(c_out),
                nn.ReLU(),
                nn.MaxPool2d(2))
        self.features = nn.Sequential(block(1, 32), block(32, 64), block(64, 128))
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Dropout(0.5),
            nn.Linear(128 * 3 * 3, 128),
            nn.ReLU(),
            nn.Dropout(0.5),
            nn.Linear(128, 10))

    def forward(self, x):
        return self.classifier(self.features(x))

torch.manual_seed(0)
cnn = ModernCNN().to(device)
print(f'Modern CNN parameters: {sum(p.numel() for p in cnn.parameters()):,}')

history_cnn = fit(cnn, epochs=10)
print(f"\nModern CNN after 10 epochs: training accuracy {history_cnn['train_acc'][-1]:.2f}%, "
      f"test accuracy {history_cnn['test_acc'][-1]:.2f}%")
```

```
Modern CNN parameters: 241,770
Epoch 1/10: train loss 0.2100, train acc 93.56% | test loss 0.0489, test acc
98.38% (7.0s)
Epoch 2/10: train loss 0.0832, train acc 97.61% | test loss 0.0281, test acc
99.10% (5.8s)
Epoch 3/10: train loss 0.0660, train acc 98.10% | test loss 0.0236, test acc
99.23% (7.6s)
Epoch 4/10: train loss 0.0567, train acc 98.39% | test loss 0.0228, test acc
99.24% (7.1s)
Epoch 5/10: train loss 0.0500, train acc 98.56% | test loss 0.0230, test acc
99.26% (7.9s)
Epoch 6/10: train loss 0.0439, train acc 98.71% | test loss 0.0218, test acc
99.19% (8.1s)
Epoch 7/10: train loss 0.0384, train acc 98.86% | test loss 0.0247, test acc
99.20% (8.4s)
Epoch 8/10: train loss 0.0377, train acc 98.87% | test loss 0.0216, test acc
99.32% (8.4s)
Epoch 9/10: train loss 0.0342, train acc 98.98% | test loss 0.0231, test acc
```

```
99.39% (8.3s)
Epoch 10/10: train loss 0.0313, train acc 99.06% | test loss 0.0219, test acc
99.34% (8.2s)

Modern CNN after 10 epochs: training accuracy 99.06%, test accuracy 99.34%
```

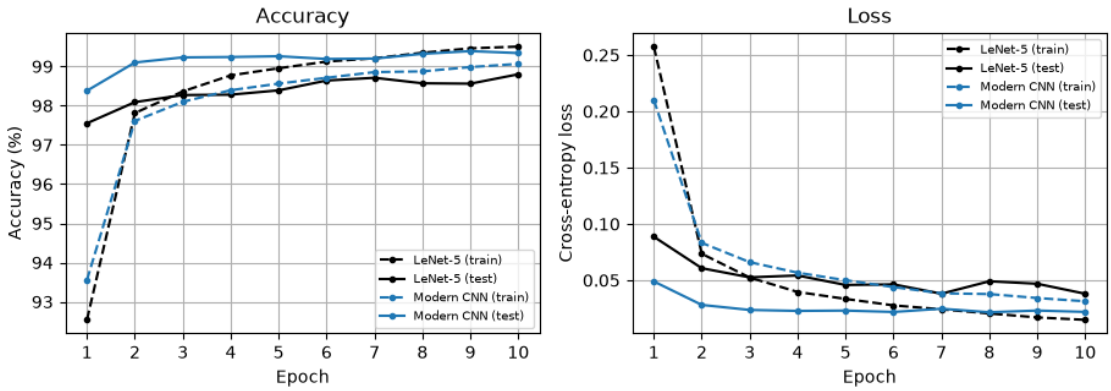
(c) Learning curves and confusion matrices

I plotted the per-epoch accuracy and loss that `fit` logged for both models, with the training values as dashed lines and the test values as solid lines.

[7]:
```
epochs = range(1, 11)
models = [(history_lenet, 'LeNet-5', 'k'), (history_cnn, 'Modern CNN', 'tab:blue')]

fig, axes = plt.subplots(1, 2, figsize=(9.3, 3.4))
for hist, name, color in models:
    axes[0].plot(epochs, hist['train_acc'], '--o', color=color, markersize=3, label=f'{name}␣
 ↪(train)')
    axes[0].plot(epochs, hist['test_acc'], '-o', color=color, markersize=3, label=f'{name}␣
 ↪(test)')
    axes[1].plot(epochs, hist['train_loss'], '--o', color=color, markersize=3, label=f'{name}␣
 ↪(train)')
    axes[1].plot(epochs, hist['test_loss'], '-o', color=color, markersize=3, label=f'{name}␣
 ↪(test)')

axes[0].set_title('Accuracy')
axes[0].set_ylabel('Accuracy (%)')
axes[1].set_title('Loss')
axes[1].set_ylabel('Cross-entropy loss')
for ax in axes:
    ax.set_xlabel('Epoch')
    ax.set_xticks(epochs)
    ax.grid(True)
    ax.legend(fontsize=7)
plt.tight_layout()
plt.show()
```



Next, I computed the confusion matrix of each model on the 10,000 test images. Row i, column j counts the test images of digit i that the model classified as j, so the correct predictions lie on the diagonal and every off-diagonal entry is a mistake.

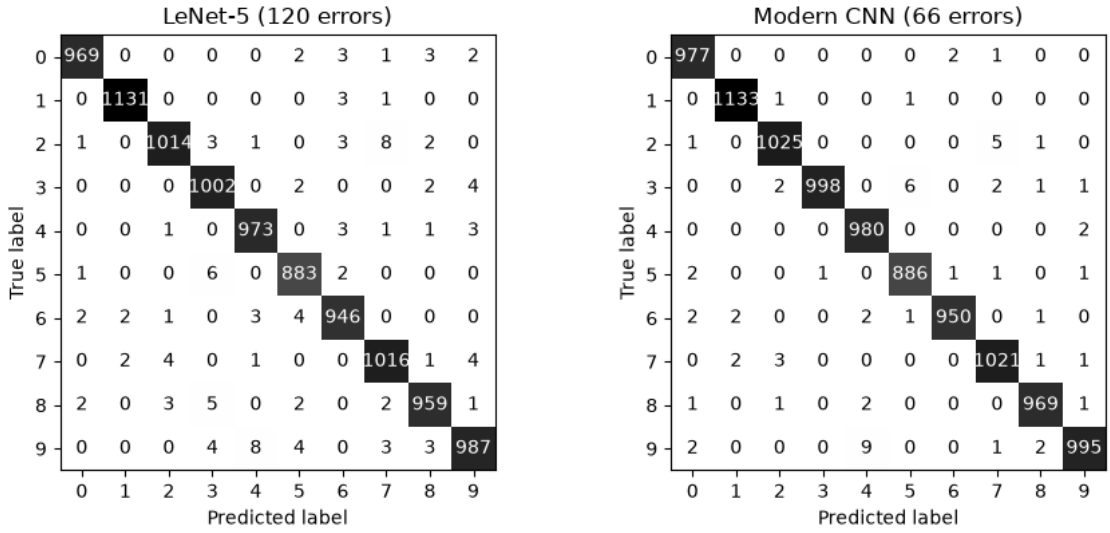
[8]:
```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

def predict(model, X, batch=1000):
    model.eval()
    with torch.no_grad():
        return torch.cat([model(X[i:i + batch]).argmax(dim=1) for i in range(0, len(X), batch)])

y_true = Ym_test.cpu().numpy()
fig, axes = plt.subplots(1, 2, figsize=(9.3, 4.2))
for ax, (model, name) in zip(axes, [(lenet, 'LeNet-5'), (cnn, 'Modern CNN')]):
    y_pred = predict(model, Xm_test).cpu().numpy()
    cm = confusion_matrix(y_true, y_pred)
    ConfusionMatrixDisplay(cm, display_labels=range(10)).plot(ax=ax, cmap='Greys',␣
 ↪colorbar=False)
    errors = cm.sum() - cm.trace()
    ax.set_title(f'{name} ({errors} errors)')
    print(f'{name}: {errors} misclassified test images')

    # Most frequent mistake (largest off-diagonal entry)
    off = cm.copy()
    off[list(range(10)), list(range(10))] = 0
    i, j = divmod(off.argmax(), 10)
    print(f'  most common mistake: {i} predicted as {j} ({off[i, j]} times)')
plt.tight_layout()
plt.show()
```

```
LeNet-5: 120 misclassified test images
  most common mistake: 2 predicted as 7 (8 times)
Modern CNN: 66 misclassified test images
  most common mistake: 9 predicted as 4 (9 times)
```



Discussion: both networks reached a high accuracy on MNIST within 10 epochs, but the modern CNN was clearly better. LeNet-5 finished with 99.50% training and 98.80% test accuracy (120 misclassified test images), while my modern CNN finished with 99.06% training and 99.34% test accuracy (66 errors), so it made about 45% fewer mistakes on the same test set. It was also better

from the very first epoch (98.38% test accuracy after one epoch against 97.55% for LeNet-5), which I think comes from batch normalization and ReLU: batch normalization keeps the inputs of each layer well scaled, and ReLU does not saturate like tanh, so the gradients stay large and the first epoch already makes a lot of progress.

The learning curves show a difference in how the two models generalize. For LeNet-5 the training accuracy passed the test accuracy from epoch 3 onwards, and its test loss stopped improving and started to fluctuate after epoch 7 (between 0.038 and 0.049) even though the training loss kept falling, which is the beginning of overfitting. For the modern CNN the training accuracy stayed *below* the test accuracy for all 10 epochs. This does not mean the model is worse on the training data: the training accuracy is measured while dropout is switching off half of the classifier units, whereas the test accuracy is measured with all units active. Together with its stable test loss (around 0.022 from epoch 4), this shows that dropout prevented overfitting even though the modern CNN has about four times as many parameters (241,770 against 61,706).

The confusion matrices show that both models make the same kind of mistakes, between digits that look alike when written by hand. For LeNet-5 the most common errors were 2 predicted as 7 and 9 predicted as 4 (8 times each), followed by 5 predicted as 3. The modern CNN removed most of these, and its remaining errors are spread thinly over the matrix; its most common mistake was 9 predicted as 4 (9 times), which are the hardest pair because a 4 with a closed top looks very similar to a 9. The price of the better accuracy is computation: each epoch of the modern CNN took about 8 s compared with about 5 s for LeNet-5 on my GPU.

Question 3

The hymenoptera dataset has images of ants and bees, with only about 120 training images and 75 validation images per class. This is far too little to train a network like ResNet18 from scratch, so I started from a ResNet18 that was pre-trained on ImageNet-1K (1.2 million images, 1000 classes), following the PyTorch transfer learning tutorial.

I used the same preprocessing as the tutorial. The training images are augmented with a random resized crop to 224×224 and a random horizontal flip, so that the network sees slightly different versions of each image in every epoch. The validation images are resized to 256 and center-cropped to 224 without any randomness. Both are normalized with the ImageNet mean and standard deviation, because that is what the pre-trained weights expect.

[9]:
```
import copy
import time
import torch.optim as optim
import numpy as np
from torchvision import datasets, models
from torchvision.models import ResNet18_Weights

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')

imagenet_mean, imagenet_std = [0.485, 0.456, 0.406], [0.229, 0.224, 0.225]
data_transforms = {
    'train': transforms.Compose([
        transforms.RandomResizedCrop(224),
        transforms.RandomHorizontalFlip(),
        transforms.ToTensor(),
        transforms.Normalize(imagenet_mean, imagenet_std)]),
    'val': transforms.Compose([
```

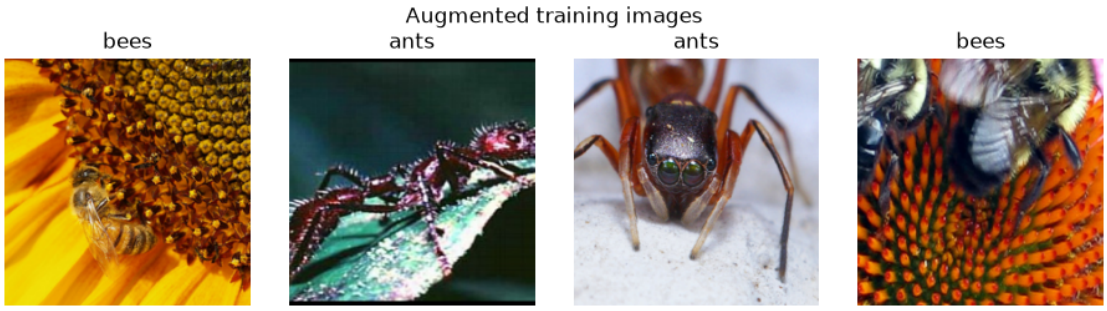
```
        transforms.Resize(256),
        transforms.CenterCrop(224),
        transforms.ToTensor(),
        transforms.Normalize(imagenet_mean, imagenet_std)]),
}

data_dir = './data/hymenoptera_data'
image_datasets = {x: datasets.ImageFolder(f'{data_dir}/{x}', data_transforms[x]) for x in␣
↪['train', 'val']}
dataloaders = {x: torch.utils.data.DataLoader(image_datasets[x], batch_size=4, shuffle=(x ==␣
↪'train'), num_workers=0)
               for x in ['train', 'val']}
dataset_sizes = {x: len(image_datasets[x]) for x in ['train', 'val']}
class_names = image_datasets['train'].classes
print('Classes:', class_names)
print('Images:', dataset_sizes)
```

```
Classes: ['ants', 'bees']
Images: {'train': 244, 'val': 153}
```

[10]:
```
def show_tensor(ax, img, title):
    # Undo the normalization so the image can be displayed
    img = img.numpy().transpose((1, 2, 0)) * np.array(imagenet_std) + np.array(imagenet_mean)
    ax.imshow(np.clip(img, 0, 1))
    ax.set_title(title)
    ax.axis('off')

torch.manual_seed(0)
inputs, labels = next(iter(dataloaders['train']))
fig, axes = plt.subplots(1, 4, figsize=(9.3, 2.6))
for ax, img, lab in zip(axes, inputs, labels):
    show_tensor(ax, img, class_names[lab])
plt.suptitle('Augmented training images')
plt.tight_layout()
plt.show()
```



I wrote one training function and use it for both parts, so the two approaches are trained under identical conditions: SGD with momentum 0.9, a learning rate of 0.001 that is divided by 10 every 7 epochs, a batch size of 4 and 25 epochs, as in the tutorial. After every epoch it evaluates the model on the validation set and records the loss and accuracy of both phases. Like the tutorial, it keeps a copy of the weights from the epoch with the best validation accuracy and loads them back at the end.

[14]:
```
def train_model(model, optimizer, scheduler, num_epochs=25):
    criterion = nn.CrossEntropyLoss()
    history = {f'{phase}_{m}': [] for phase in ['train', 'val'] for m in ['loss', 'acc']}
    best_wts, best_acc = copy.deepcopy(model.state_dict()), 0.0
    start = time.time()

    for epoch in range(num_epochs):
        for phase in ['train', 'val']:
            model.train() if phase == 'train' else model.eval()
            running_loss, running_corrects = 0.0, 0
            for inputs, labels in dataloaders[phase]:
                inputs, labels = inputs.to(device), labels.to(device)
                optimizer.zero_grad()
                with torch.set_grad_enabled(phase == 'train'):
                    outputs = model(inputs)
                    loss = criterion(outputs, labels)
                    if phase == 'train':
                        loss.backward()
                        optimizer.step()
                running_loss += loss.item() * inputs.size(0)
                running_corrects += (outputs.argmax(dim=1) == labels).sum().item()
            if phase == 'train':
                scheduler.step()

            history[f'{phase}_loss'].append(running_loss / dataset_sizes[phase])
            history[f'{phase}_acc'].append(100 * running_corrects / dataset_sizes[phase])
            if phase == 'val' and history['val_acc'][-1] > best_acc:
                best_acc = history['val_acc'][-1]
                best_wts = copy.deepcopy(model.state_dict())

        print(f"Epoch {epoch + 1}/{num_epochs}: "
              f"train loss {history['train_loss'][-1]:.4f}, acc {history['train_acc'][-1]:.2f}%␣
↪| "
              f"val loss {history['val_loss'][-1]:.4f}, acc {history['val_acc'][-1]:.2f}%")

    print(f'Training time: {time.time() - start:.0f}s, best validation accuracy: {best_acc:.
↪2f}%')
    model.load_state_dict(best_wts)
    return model, history, best_acc
```

(a) Fine-tuning

For fine-tuning, I loaded the pre-trained ResNet18 and replaced its last fully connected layer (512 inputs, 1000 ImageNet classes) with a new layer that has 2 outputs, one for ants and one for bees. Then I trained the whole network, so all the pre-trained weights are updated together with the new layer. The small learning rate (0.001) is important here: it lets the network adapt its features to ants and bees without destroying what it already learned from ImageNet.

[15]:
```
torch.manual_seed(0)
model_ft = models.resnet18(weights=ResNet18_Weights.IMAGENET1K_V1)
model_ft.fc = nn.Linear(model_ft.fc.in_features, len(class_names))
model_ft = model_ft.to(device)

trainable = sum(p.numel() for p in model_ft.parameters() if p.requires_grad)
print(f'Trainable parameters: {trainable:,}')

optimizer_ft = optim.SGD(model_ft.parameters(), lr=0.001, momentum=0.9)
```

```
scheduler_ft = optim.lr_scheduler.StepLR(optimizer_ft, step_size=7, gamma=0.1)
model_ft, history_ft, best_acc_ft = train_model(model_ft, optimizer_ft, scheduler_ft,␣
↪num_epochs=25)
```

```
Trainable parameters: 11,177,538
Epoch 1/25: train loss 0.5849, acc 68.85% | val loss 0.2434, acc 91.50%
Epoch 2/25: train loss 0.5221, acc 77.05% | val loss 0.2342, acc 89.54%
Epoch 3/25: train loss 0.6879, acc 70.08% | val loss 0.3038, acc 89.54%
Epoch 4/25: train loss 0.4691, acc 80.74% | val loss 0.2478, acc 88.89%
Epoch 5/25: train loss 0.4901, acc 78.69% | val loss 0.2694, acc 88.89%
Epoch 6/25: train loss 0.2945, acc 88.52% | val loss 0.1666, acc 96.08%
Epoch 7/25: train loss 0.4400, acc 82.38% | val loss 0.2364, acc 90.20%
Epoch 8/25: train loss 0.3622, acc 82.79% | val loss 0.1833, acc 92.16%
Epoch 9/25: train loss 0.3623, acc 86.07% | val loss 0.2185, acc 90.20%
Epoch 10/25: train loss 0.3153, acc 87.30% | val loss 0.1843, acc 90.85%
Epoch 11/25: train loss 0.3781, acc 86.07% | val loss 0.1977, acc 92.16%
Epoch 12/25: train loss 0.2593, acc 85.66% | val loss 0.1696, acc 92.16%
Epoch 13/25: train loss 0.3301, acc 86.89% | val loss 0.1963, acc 90.20%
Epoch 14/25: train loss 0.2979, acc 85.25% | val loss 0.2197, acc 91.50%
Epoch 15/25: train loss 0.2725, acc 89.34% | val loss 0.2967, acc 89.54%
Epoch 16/25: train loss 0.2480, acc 88.93% | val loss 0.2016, acc 91.50%
Epoch 17/25: train loss 0.2781, acc 88.52% | val loss 0.2283, acc 90.20%
Epoch 18/25: train loss 0.3207, acc 85.25% | val loss 0.2508, acc 90.85%
Epoch 19/25: train loss 0.1725, acc 93.44% | val loss 0.2126, acc 90.85%
Epoch 20/25: train loss 0.2596, acc 88.93% | val loss 0.1944, acc 90.85%
Epoch 21/25: train loss 0.3002, acc 85.66% | val loss 0.2150, acc 90.85%
Epoch 22/25: train loss 0.2807, acc 86.89% | val loss 0.1795, acc 90.20%
Epoch 23/25: train loss 0.1889, acc 93.85% | val loss 0.1940, acc 91.50%
Epoch 24/25: train loss 0.3148, acc 87.30% | val loss 0.1879, acc 90.85%
Epoch 25/25: train loss 0.2379, acc 91.39% | val loss 0.1851, acc 92.16%
Training time: 108s, best validation accuracy: 96.08%
```

(b) Using the network as a feature extractor

For the second approach, I loaded a fresh pre-trained ResNet18 and froze all of its layers by setting `requires_grad = False`, so the convolutional layers keep exactly the weights learned on ImageNet. Then I replaced the last layer with a new 2-output layer, which is trainable by default, and gave only this layer's parameters to the optimizer. The network therefore works as a fixed feature extractor: every image is turned into the same 512-dimensional feature vector as before, and only a linear classifier on top of these features is learned.

Since no gradients have to be computed through the frozen layers, training is much cheaper. Everything else (optimizer, learning rate schedule, batch size and 25 epochs) is the same as in part (a).

[16]:
```
torch.manual_seed(0)
model_fe = models.resnet18(weights=ResNet18_Weights.IMAGENET1K_V1)
for param in model_fe.parameters():
    param.requires_grad = False  # freeze the pre-trained layers

model_fe.fc = nn.Linear(model_fe.fc.in_features, len(class_names))  # new layer, trainable
model_fe = model_fe.to(device)

trainable = sum(p.numel() for p in model_fe.parameters() if p.requires_grad)
print(f'Trainable parameters: {trainable:,}')

optimizer_fe = optim.SGD(model_fe.fc.parameters(), lr=0.001, momentum=0.9)
```

```
scheduler_fe = optim.lr_scheduler.StepLR(optimizer_fe, step_size=7, gamma=0.1)
model_fe, history_fe, best_acc_fe = train_model(model_fe, optimizer_fe, scheduler_fe,␣
 ↪num_epochs=25)
```

```
Trainable parameters: 1,026
Epoch 1/25: train loss 0.6504, acc 63.93% | val loss 0.2278, acc 90.85%
Epoch 2/25: train loss 0.5138, acc 75.82% | val loss 0.1872, acc 95.42%
Epoch 3/25: train loss 0.6806, acc 70.08% | val loss 0.1718, acc 94.77%
Epoch 4/25: train loss 0.4656, acc 81.97% | val loss 0.1525, acc 96.08%
Epoch 5/25: train loss 0.4344, acc 79.92% | val loss 0.2104, acc 92.81%
Epoch 6/25: train loss 0.3057, acc 85.25% | val loss 0.1705, acc 95.42%
Epoch 7/25: train loss 0.3711, acc 82.38% | val loss 0.1670, acc 95.42%
Epoch 8/25: train loss 0.3751, acc 84.43% | val loss 0.1589, acc 95.42%
Epoch 9/25: train loss 0.4059, acc 80.33% | val loss 0.1724, acc 94.77%
Epoch 10/25: train loss 0.3722, acc 85.25% | val loss 0.1539, acc 95.42%
Epoch 11/25: train loss 0.4110, acc 81.97% | val loss 0.1569, acc 94.12%
Epoch 12/25: train loss 0.3158, acc 88.11% | val loss 0.1595, acc 95.42%
Epoch 13/25: train loss 0.3722, acc 82.38% | val loss 0.1547, acc 95.42%
Epoch 14/25: train loss 0.3270, acc 82.79% | val loss 0.1867, acc 92.81%
Epoch 15/25: train loss 0.2991, acc 87.30% | val loss 0.2152, acc 92.81%
Epoch 16/25: train loss 0.3018, acc 86.48% | val loss 0.1643, acc 94.77%
Epoch 17/25: train loss 0.2955, acc 86.07% | val loss 0.1712, acc 94.77%
Epoch 18/25: train loss 0.3937, acc 81.97% | val loss 0.2020, acc 92.81%
Epoch 19/25: train loss 0.2339, acc 89.75% | val loss 0.1604, acc 94.77%
Epoch 20/25: train loss 0.3071, acc 87.70% | val loss 0.1643, acc 95.42%
Epoch 21/25: train loss 0.3672, acc 84.02% | val loss 0.1763, acc 94.77%
Epoch 22/25: train loss 0.3269, acc 83.61% | val loss 0.1587, acc 96.08%
Epoch 23/25: train loss 0.2447, acc 90.16% | val loss 0.1721, acc 94.77%
Epoch 24/25: train loss 0.3394, acc 83.20% | val loss 0.1588, acc 94.12%
Epoch 25/25: train loss 0.3143, acc 88.11% | val loss 0.1695, acc 94.77%
Training time: 107s, best validation accuracy: 96.08%
```